

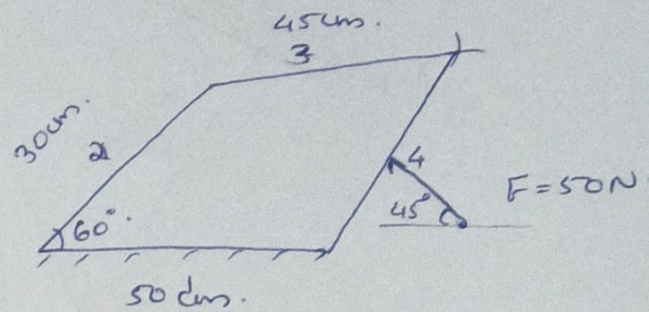
## Force analysis.

1). Static force analysis

$$\Sigma F = 0$$

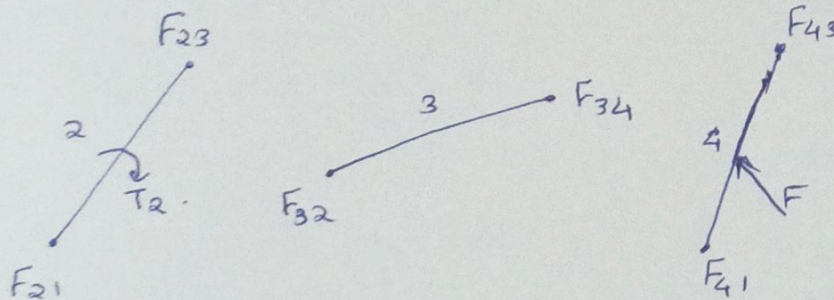
2). Dynamic force analysis.

$$\Sigma F = m \cdot a \rightarrow \text{acc}^n.$$



Grashoff's criteria  $\rightarrow s + l < a + b$ .

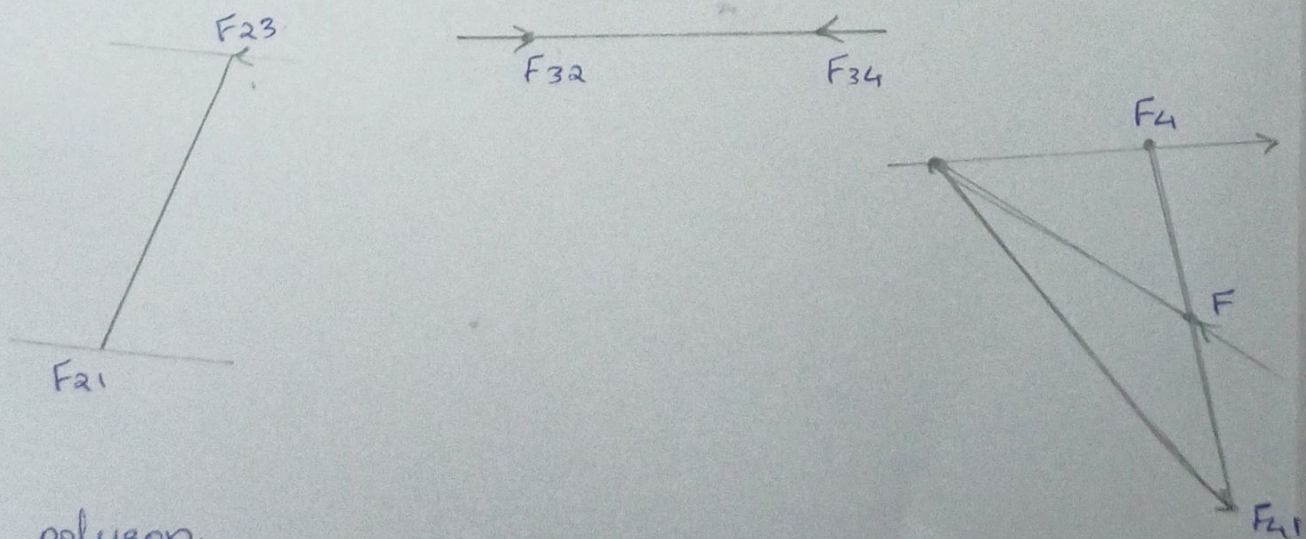
short link      longest link      other links.



Link 2: 2 force torque.

Link 3: 2 force member [equilibrium if equal, oppo, collinear]

Link 4: 3 force member [equilibrium if only they are concurrent]



## Force polygon.

3 forces in equilibrium should form a closed polygon  $\Delta^{\text{ce}}$ .



Scale  $\Rightarrow 1:10$

1 cm = 10 N.

||  $\perp$  to  $F_{43}$ .

$F = 50 \text{ N}$